

Jupyter Lab Environment Setup

Jupyter Lab Environment Setup

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If you are using the factory image we provided, manual installation and configuration are not required.

Jupyter Lab is a web-based interactive development environment that supports multiple programming languages. It provides a flexible workspace for data cleaning, visualization, machine learning modeling, and various other data science tasks.

1. Install Jupyter Lab

- Check system Python version:

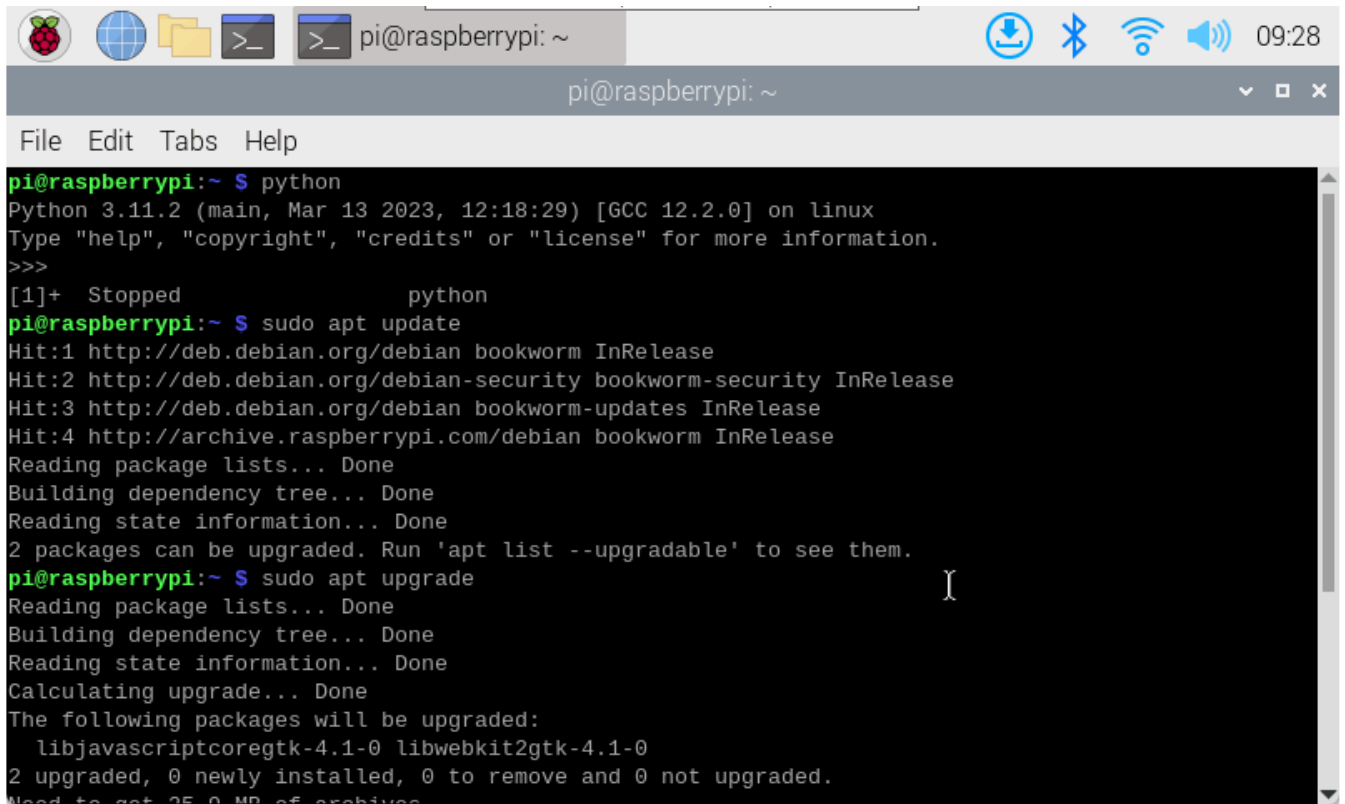
Enter the following command in the terminal:

```
python
```

- Install Jupyter Lab:

Before installing software, first update the repository list and software:

```
sudo apt update  
sudo apt upgrade
```



```
pi@raspberrypi: ~  
File Edit Tabs Help  
pi@raspberrypi:~ $ python  
Python 3.11.2 (main, Mar 13 2023, 12:18:29) [GCC 12.2.0] on linux  
Type "help", "copyright", "credits" or "license" for more information.  
>>>  
[1]+  Stopped                  python  
pi@raspberrypi:~ $ sudo apt update  
Hit:1 http://deb.debian.org/debian bookworm InRelease  
Hit:2 http://deb.debian.org/debian-security bookworm-security InRelease  
Hit:3 http://deb.debian.org/debian bookworm-updates InRelease  
Hit:4 http://archive.raspberrypi.com/debian bookworm InRelease  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
2 packages can be upgraded. Run 'apt list --upgradable' to see them.  
pi@raspberrypi:~ $ sudo apt upgrade  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
Calculating upgrade... Done  
The following packages will be upgraded:  
  libjavascriptcoregtk-4.1-0 libwebkit2gtk-4.1-0  
2 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.  
Need to get 25.0 MB of archives.
```

Install Jupyter Lab in the Python 3 environment. Enter the following command in the terminal:

```
sudo pip3 install jupyterlab
```

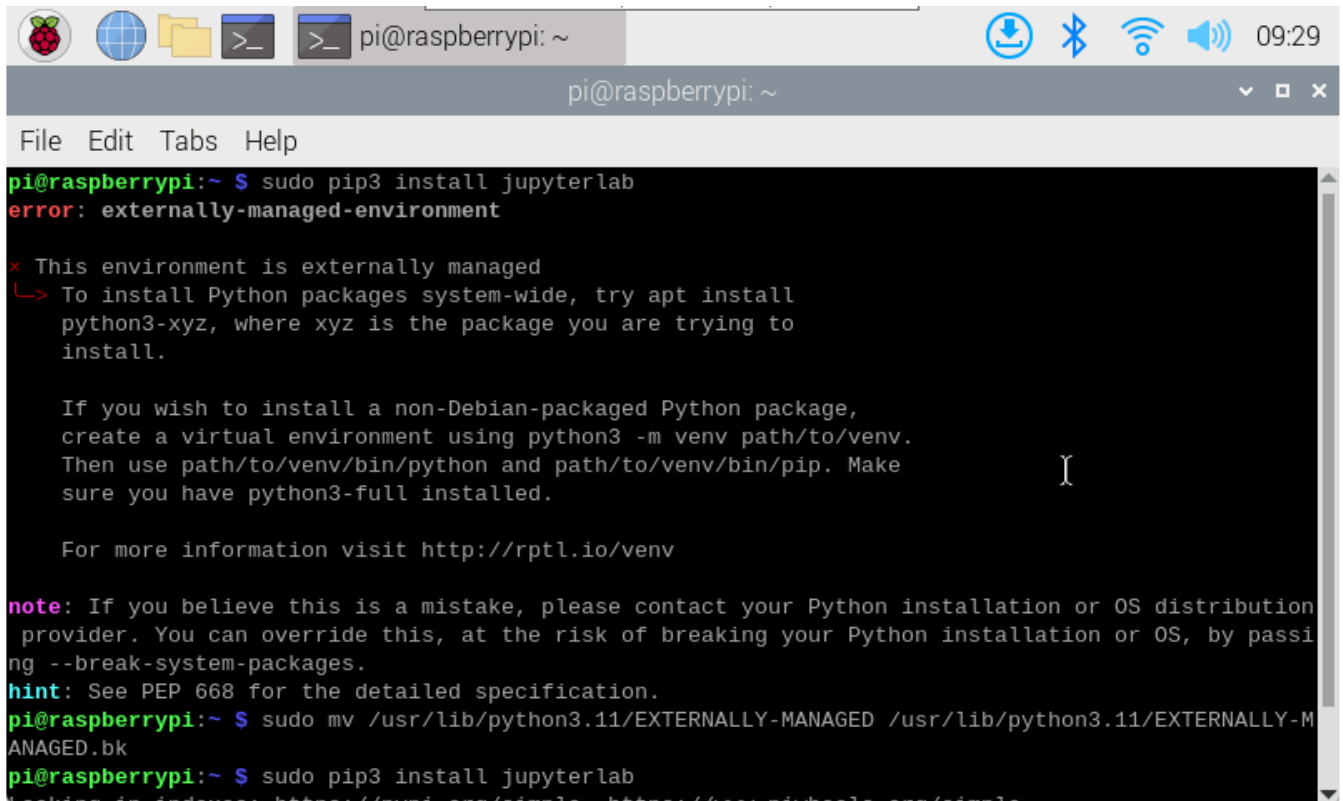
If download fails multiple times, you can specify the Tsinghua University Python package mirror address to speed up domestic downloads:

```
sudo pip3 install jupyterlab -i https://pypi.tuna.tsinghua.edu.cn/simple
```

- Error Resolution:

Entering the Jupyter Lab installation command directly in the terminal will result in the "error: externally-managed-environment" error. You can solve this using the following command. Modify the Python version according to your system version. My current system version is 3.11:

```
sudo mv /usr/lib/python3.11/EXTERNALLY-MANAGED /usr/lib/python3.11/EXTERNALLY-MANAGED.bk
```



```
pi@raspberrypi:~ $ sudo pip3 install jupyterlab
error: externally-managed-environment

× This environment is externally managed
╔> To install Python packages system-wide, try apt install
python3-xyz, where xyz is the package you are trying to
install.

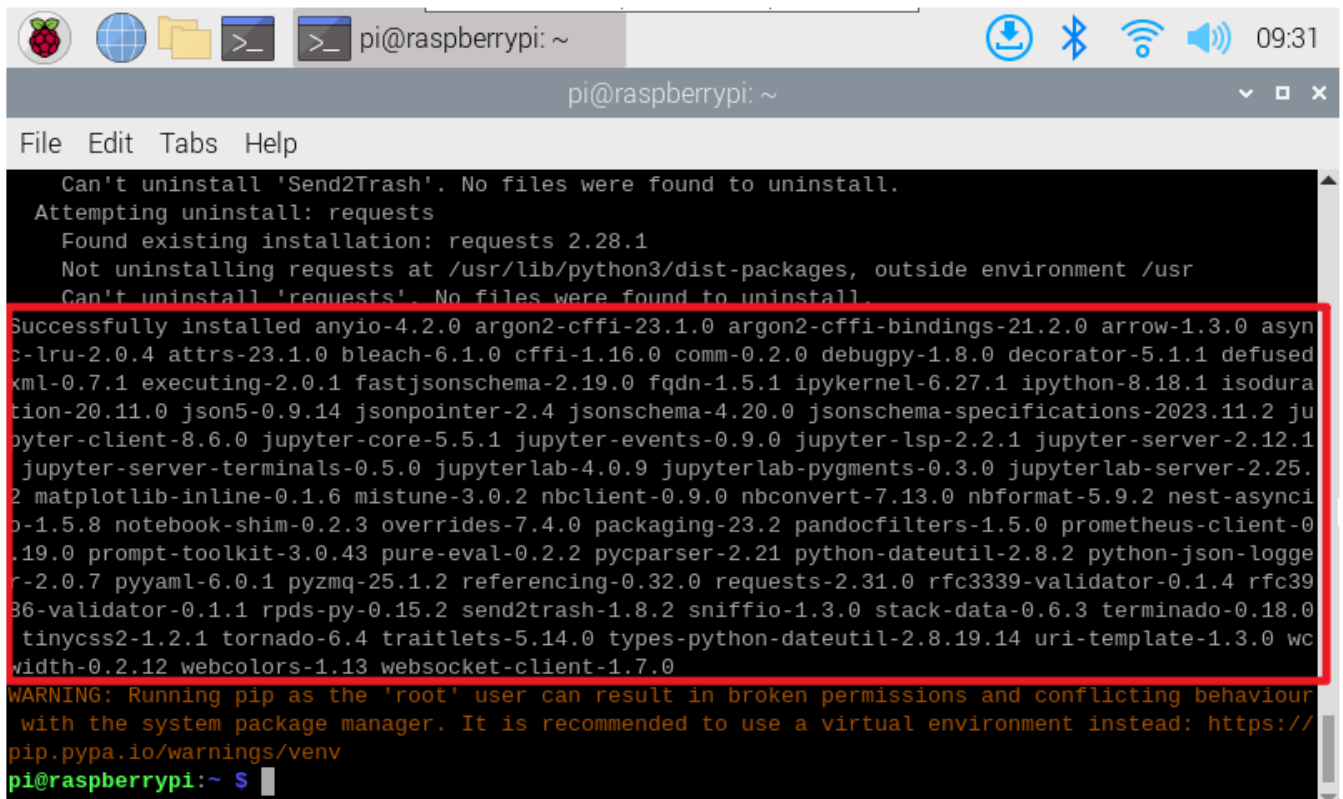
If you wish to install a non-Debian-packaged Python package,
create a virtual environment using python3 -m venv path/to/venv.
Then use path/to/venv/bin/python and path/to/venv/bin/pip. Make
sure you have python3-full installed.

For more information visit http://rptl.io/venv

note: If you believe this is a mistake, please contact your Python installation or OS distribution
provider. You can override this, at the risk of breaking your Python installation or OS, by passi
ng --break-system-packages.
hint: See PEP 668 for the detailed specification.
pi@raspberrypi:~ $ sudo mv /usr/lib/python3.11/EXTERNALLY-MANAGED /usr/lib/python3.11/EXTERNALLY-M
ANAGED.bk
pi@raspberrypi:~ $ sudo pip3 install jupyterlab
Looking in indexes: https://pypi.org/simple, https://www.pi-hole.org/simple
```

- Success Message:

The following message indicates successful installation:

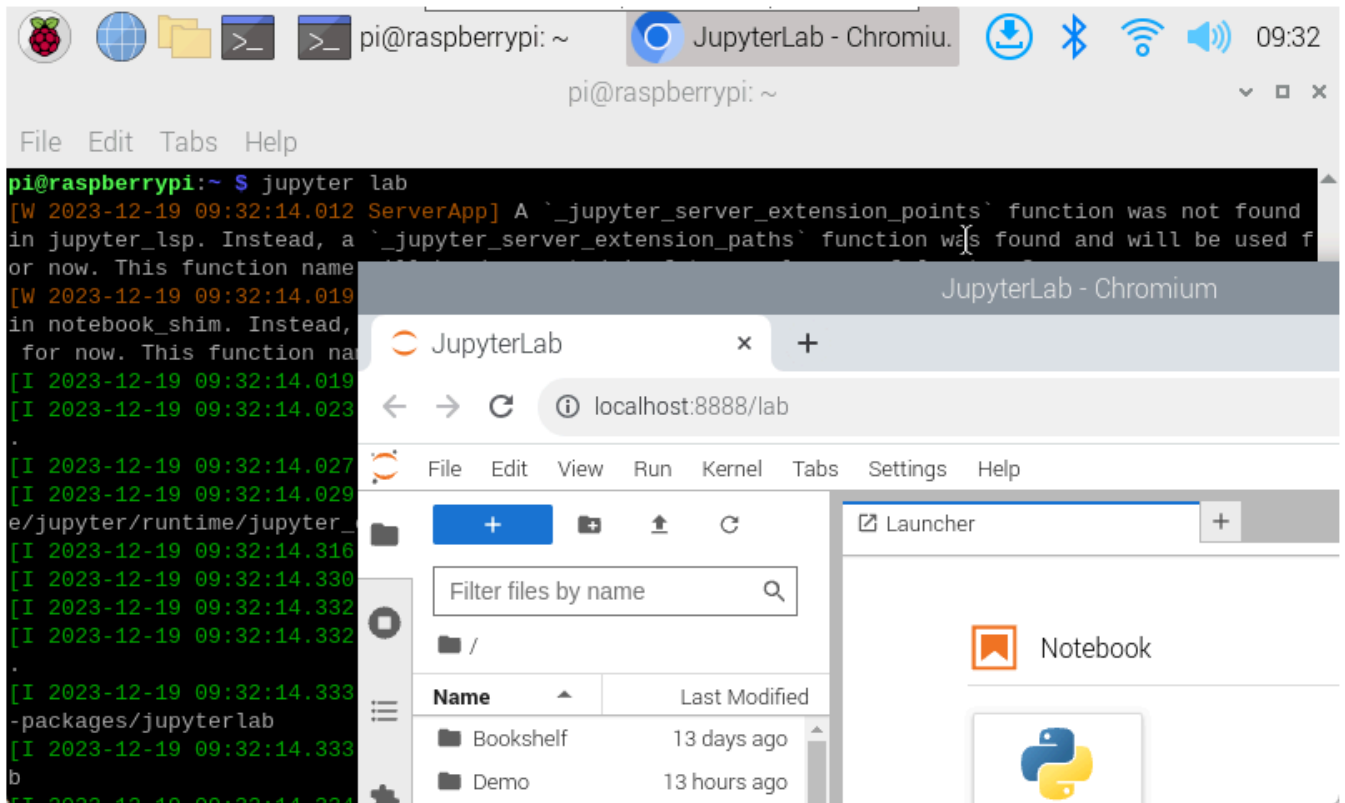


```
Can't uninstall 'Send2Trash'. No files were found to uninstall.
Attempting uninstall: requests
Found existing installation: requests 2.28.1
Not uninstalling requests at /usr/lib/python3/dist-packages, outside environment /usr
Can't uninstall 'requests'. No files were found to uninstall.
Successfully installed anyio-4.2.0 argon2-cffi-23.1.0 argon2-cffi-bindings-21.2.0 arrow-1.3.0 asyn
c-lru-2.0.4 attrs-23.1.0 bleach-6.1.0 cffi-1.16.0 comm-0.2.0 debugpy-1.8.0 decorator-5.1.1 defused
xml-0.7.1 executing-2.0.1 fastjsonschema-2.19.0 fqdn-1.5.1 ipykernel-6.27.1 ipython-8.18.1 isodura
tion-20.11.0 json5-0.9.14 jsonpointer-2.4 jsonschema-4.20.0 jsonschema-specifications-2023.11.2 ju
pyter-client-8.6.0 jupyter-core-5.5.1 jupyter-events-0.9.0 jupyter-lsp-2.2.1 jupyter-server-2.12.1
jupyter-server-terminals-0.5.0 jupyterlab-4.0.9 jupyterlab-pygments-0.3.0 jupyterlab-server-2.25.
2 matplotlib-inline-0.1.6 mistune-3.0.2 nbclient-0.9.0 nbconvert-7.13.0 nbformat-5.9.2 nest-asyncio-1.5.8 notebook-shim-0.2.3 overrides-7.4.0 packaging-23.2 pandocfilters-1.5.0 prometheus-client-0
.19.0 prompt-toolkit-3.0.43 pure-eval-0.2.2 pycparser-2.21 python-dateutil-2.8.2 python-json-logge
r-2.0.7 pyyaml-6.0.1 pyzmq-25.1.2 referencing-0.32.0 requests-2.31.0 rfc3339-validator-0.1.4 rfc39
86-validator-0.1.1 rpds-py-0.15.2 send2trash-1.8.2 sniffio-1.3.0 stack-data-0.6.3 terminado-0.18.0
tinycss2-1.2.1 tornado-6.4 traitlets-5.14.0 types-python-dateutil-2.8.19.14 uri-template-1.3.0 wc
width-0.2.12 webcolors-1.13 websocket-client-1.7.0
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour
with the system package manager. It is recommended to use a virtual environment instead: https://
pip.pypa.io/warnings/venv
pi@raspberrypi:~ $
```

2. Open Jupyter Lab

Enter `jupyter lab` in the terminal. If a password is required, follow step 4 in the tutorial to set the password first before using!

```
jupyter lab
```



Before installing Jupyter Lab, select the system's default browser first. Otherwise, Jupyter Lab will not start directly from the browser.

When using the sudo command for Jupyter Lab installation, the warning message can be ignored.

3. Set Up LAN Access for Jupyter Lab

- Create configuration file:

The generated configuration file path is the path to modify:

```
jupyter lab --generate-config
```

```
pi@raspberrypi:~ $ jupyter lab --generate-config
Writing default config to: /home/pi/.jupyter/jupyter_lab_config.py
```

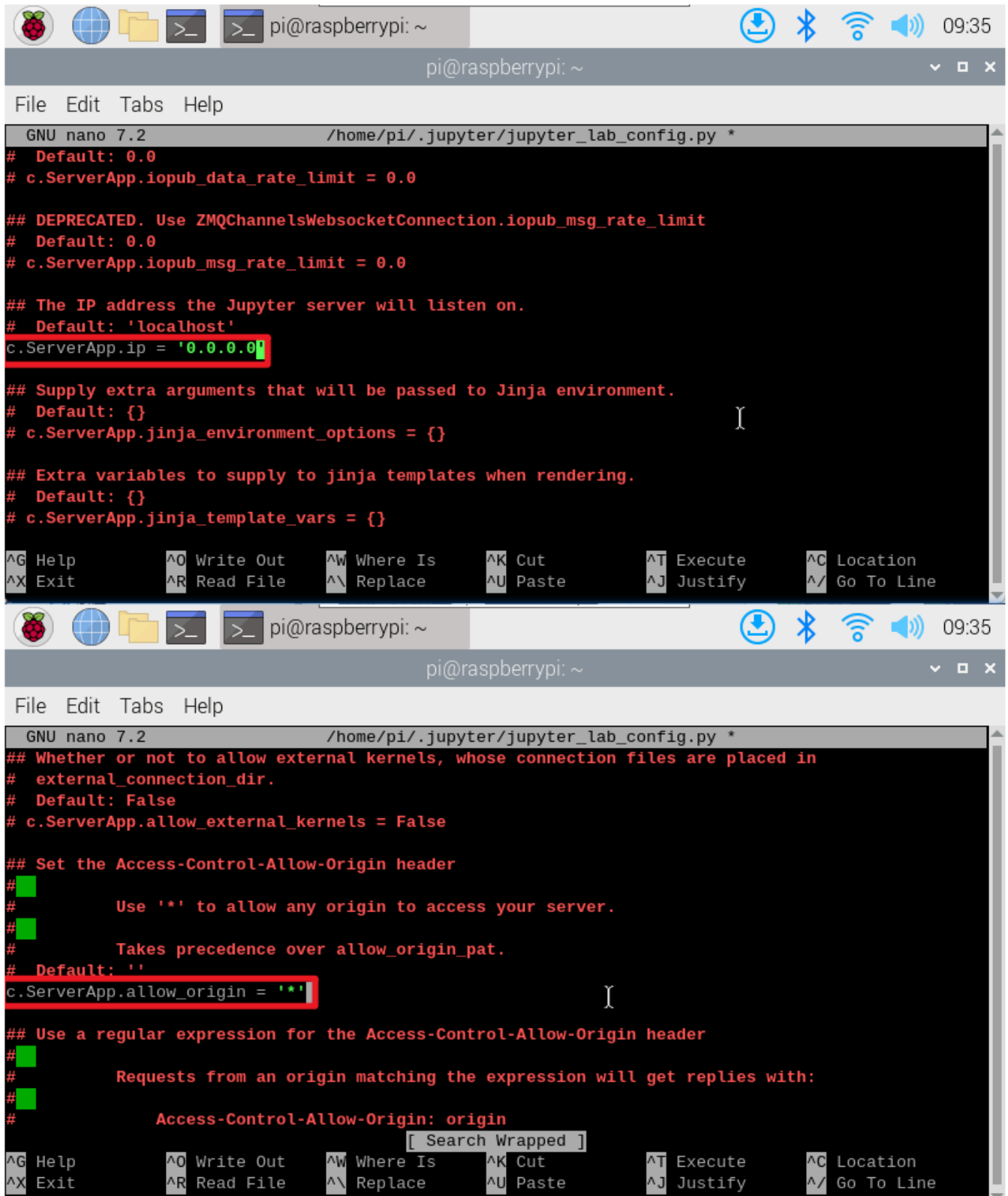
- Modify configuration file:

```
sudo nano /home/pi/.jupyter/jupyter_lab_config.py
```

Remove comments from the file content and modify to the following. In the nano editor, you can use Ctrl+W shortcut to search for keywords:

```
c.ServerApp.allow_origin = '*'
c.ServerApp.ip = '0.0.0.0'
```

Press Ctrl+X, enter Y, then press Enter to save and exit!



The image shows two screenshots of a terminal window on a Raspberry Pi, displaying the configuration file `/home/pi/.jupyter/jupyter_lab_config.py` using the `nano` text editor. The terminal window has a title bar with the Raspberry Pi logo, window icons, and the prompt `pi@raspberrypi: ~`. The top status bar shows the time as 09:35 and various system icons.

Top Screenshot: The configuration file is open, and the line `c.ServerApp.ip = '0.0.0.0'` is highlighted with a red box. The cursor is at the end of this line. The file content includes comments about data rate limits, deprecated settings, and the IP address the Jupyter server will listen on.

```
GNU nano 7.2 /home/pi/.jupyter/jupyter_lab_config.py *
# Default: 0.0
# c.ServerApp.iopub_data_rate_limit = 0.0

## DEPRECATED. Use ZMQChannelsWebsocketConnection.iopub_msg_rate_limit
# Default: 0.0
# c.ServerApp.iopub_msg_rate_limit = 0.0

## The IP address the Jupyter server will listen on.
# Default: 'localhost'
c.ServerApp.ip = '0.0.0.0'

## Supply extra arguments that will be passed to Jinja environment.
# Default: {}
# c.ServerApp.jinja_environment_options = {}

## Extra variables to supply to jinja templates when rendering.
# Default: {}
# c.ServerApp.jinja_template_vars = {}

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

Bottom Screenshot: The configuration file is open, and the line `c.ServerApp.allow_origin = '*'` is highlighted with a red box. The cursor is at the end of this line. The file content includes comments about external kernels, the Access-Control-Allow-Origin header, and the use of a regular expression for the header.

```
GNU nano 7.2 /home/pi/.jupyter/jupyter_lab_config.py *
## Whether or not to allow external kernels, whose connection files are placed in
# external_connection_dir.
# Default: False
# c.ServerApp.allow_external_kernels = False

## Set the Access-Control-Allow-Origin header
#
# Use '*' to allow any origin to access your server.
#
# Takes precedence over allow_origin_pat.
# Default: ''
c.ServerApp.allow_origin = '*'

## Use a regular expression for the Access-Control-Allow-Origin header
#
# Requests from an origin matching the expression will get replies with:
#
# Access-Control-Allow-Origin: origin
[ Search Wrapped ]

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

4. Set Up Access for Jupyter Lab

Enter the password setting command in the terminal. You need to enter it twice. The entered password will not be displayed:

```
jupyter lab password
```

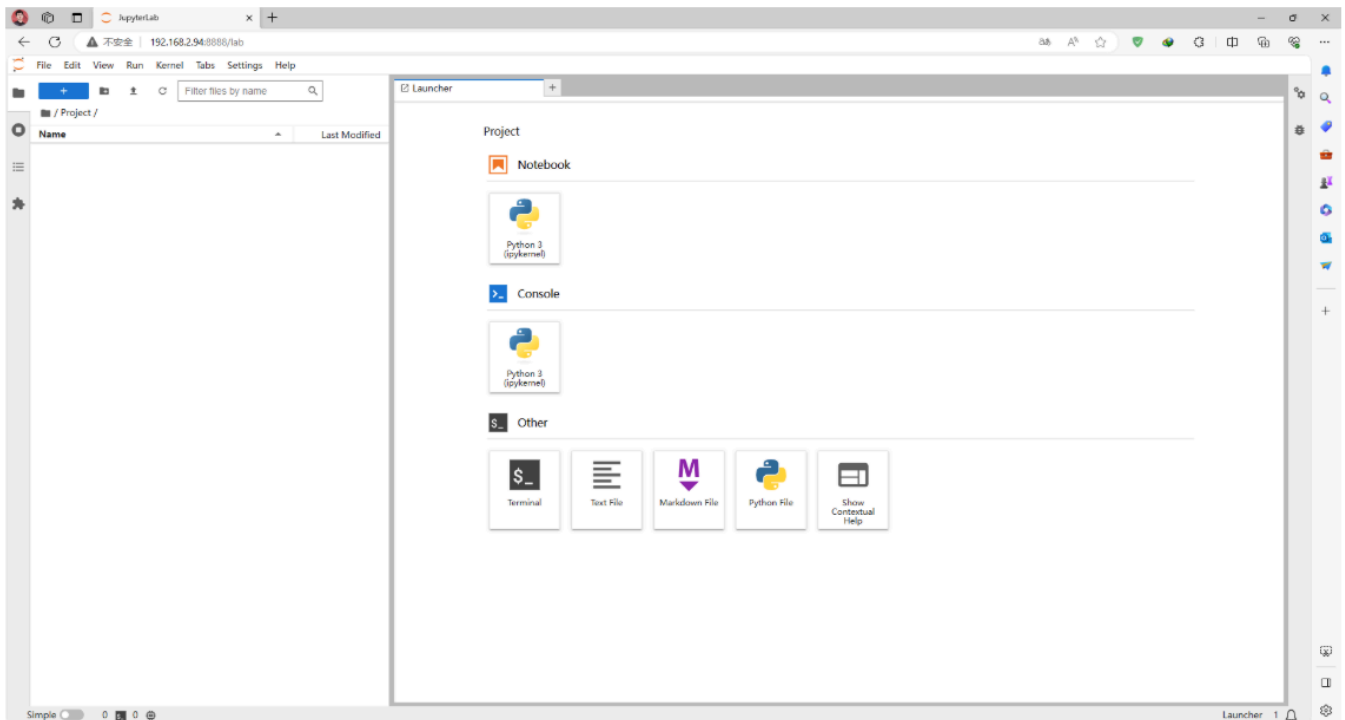
```
pi@raspberrypi:~$ jupyter lab password
Enter password:
Verify password:
[JupyterPasswordApp] Wrote hashed password to /home/pi/.jupyter/jupyter_server_config.json
```

Restart Raspberry Pi after setting the password!

- Verification:

Devices on the same LAN can access by entering IP:8888 in the browser!

The password is the one you set earlier: yahboom



5. Set Jupyter Lab to Start on Boot

After completing the above steps, you need to enter the command in the terminal to use Jupyter Lab each time. For more convenience, we can configure Jupyter Lab to start automatically on boot.

- Configure startup item:

Enter the following command in the terminal:

```
sudo nano /etc/systemd/system/jupyter.service
```

Add the following content to the file:

```
[Unit]
Description=Jupyter Lab
After=network.target
[Service]
Type=simple
ExecStart=/usr/local/bin/jupyter-lab
config=/home/pi/.jupyter/jupyter_lab_config.py --no-browser
User=pi
```

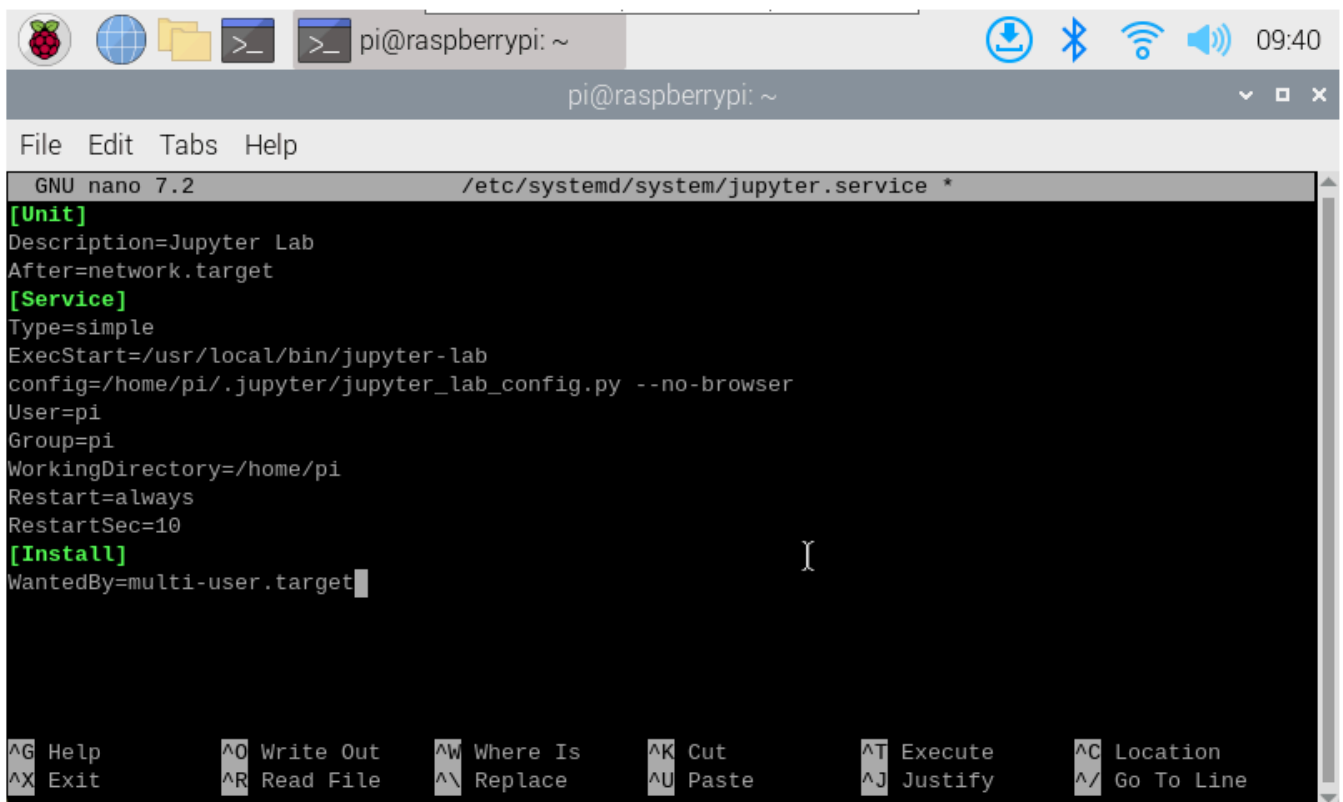
```
Group=pi
WorkingDirectory=/home/pi
Restart=always
RestartSec=10
[Install]
WantedBy=multi-user.target
```

pi: My current system's username

ExecStart: Command to start Jupyter Lab. Modify to the JupyterLab installation path and configuration file path (if you followed all the steps as instructed, you can enter the same path):

view jupyter-lab installation path: which jupyter-lab
Configuration file path reference the path generated above

WorkingDirectory: Jupyter-lab's working directory, can be changed as needed.



```
pi@raspberrypi: ~
File Edit Tabs Help
GNU nano 7.2 /etc/systemd/system/jupyter.service *
[Unit]
Description=Jupyter Lab
After=network.target
[Service]
Type=simple
ExecStart=/usr/local/bin/jupyter-lab
config=/home/pi/.jupyter/jupyter_lab_config.py --no-browser
User=pi
Group=pi
WorkingDirectory=/home/pi
Restart=always
RestartSec=10
[Install]
WantedBy=multi-user.target
^G Help  ^O Write Out  ^W Where Is  ^K Cut  ^T Execute  ^C Location
^X Exit  ^R Read File  ^_ Replace  ^U Paste  ^J Justify  ^_ Go To Line
```

- jupyter.service service:

Enable auto-start on boot:

```
sudo systemctl enable jupyter
```

Disable auto-start on boot:

```
sudo systemctl disable jupyter
```

Start service:

```
sudo systemctl start jupyter
```

Stop service:

```
sudo systemctl stop jupyter
```

Check service status:

```
sudo systemctl status jupyter
```

After entering the commands to enable jupyter.service auto-start and start the service in the terminal, restart the Raspberry Pi system.

```
pi@raspberrypi:~ $ sudo nano /etc/systemd/system/jupyter.service
pi@raspberrypi:~ $ sudo systemctl enable jupyter
Created symlink /etc/systemd/system/multi-user.target.wants/jupyter.service → /etc/systemd/system/jupyter.service.
pi@raspberrypi:~ $ sudo systemctl start jupyter
```

After completing the above steps, you can access via LAN without entering `jupyter lab` in the terminal!